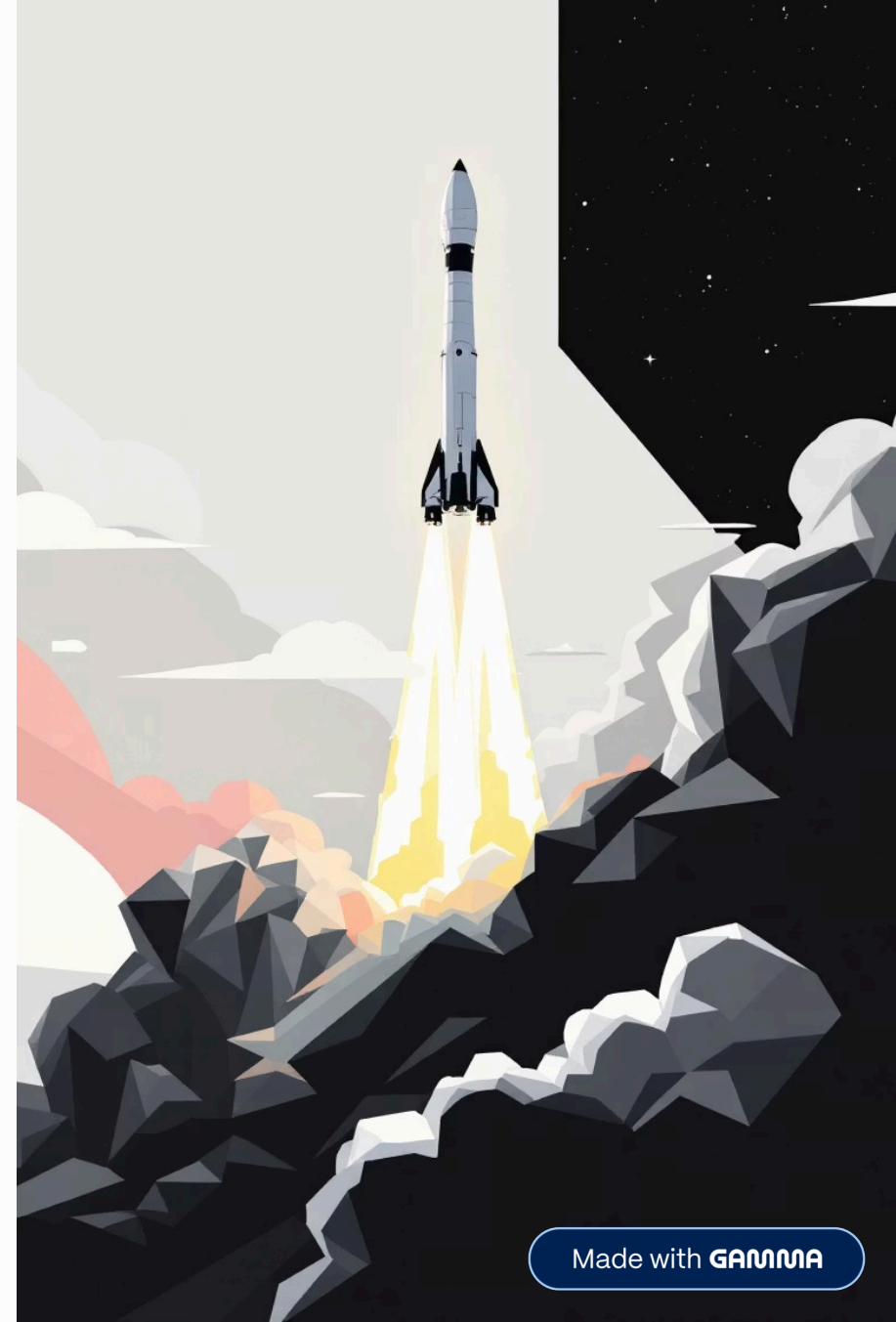


SpaceX Falcon 9 Launch Data Analysis

Data Science Capstone Project

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Made with GAMMA

Executive Summary

Project Goal

Analyse Falcon 9 launch data to understand trends, success rates, and predict mission outcomes using advanced data science techniques.

Key Achievement

Predictive models achieved up to 93% accuracy in forecasting launch success, with comprehensive insights across launch sites and payload configurations.





Critical Success Insights

80%

Overall Success Rate

Across all launch sites and mission types

93%

Model Accuracy

Best performing predictive algorithm achievement

#1

Top Launch Site

CCAFS SLC-40 leads in total launches

Payload mass and orbit type emerged as the strongest predictors of mission success, providing valuable insights for future launch planning.

Data Collection & Sources

Data Acquisition Methods

- SpaceX API GET requests for comprehensive launch data
- Wikipedia scraping using BeautifulSoup for additional metadata
- SQL database integration for structured analysis



The dataset encompasses Falcon 9 launches, payload specifications, launch site details, and comprehensive mission outcomes, creating a robust foundation for analysis.

Data Wrangling Process

01

Launch Data Filtering

Isolated Falcon 9 launches from broader SpaceX mission data

02

Missing Value Treatment

Replaced null PayloadMass values with calculated means

03

Feature Engineering

Created dummy variables for Orbit, LaunchSite, LandingPad, and Serial columns

04

Quality Validation

Verified completeness of landing outcomes and launch site data





Exploratory Data Analysis Approach



Statistical Analysis

Launch success rates per site, payload mass distributions, and orbit type frequency analysis using Pandas and NumPy.



Geographic Visualisation

Interactive Folium maps displaying launch sites with outcome-based colour coding and detailed landing information.



Interactive Dashboards

Plotly Dash interfaces enabling dynamic filtering and real-time exploration of mission outcomes and trends.

Predictive Modelling Strategy

Algorithm Selection

- Decision Tree Classifier for interpretable rules
- K-Nearest Neighbours for pattern recognition
- Support Vector Machine for complex boundaries

Methodology

- 80/20 train-test data split
- Hyperparameter optimisation
- Confusion matrix evaluation





Key Visualisation Insights

1

Temporal Trends

Annual launch frequency analysis reveals increasing mission cadence and improved success rates over time.

2

Site Performance

Launch success rates vary significantly by location, with certain sites demonstrating superior operational outcomes.

3

Payload Impact

Scatter plot analysis shows clear correlation between payload mass and mission success probability.

SQL Database Analysis Results



1

Launch Site Distribution

CCAFS SLC-40 emerged as the most active launch facility, handling the majority of Falcon 9 missions with consistent performance metrics.

2

Payload Analysis

SQL queries revealed significant variations in payload mass, identifying optimal ranges for successful mission outcomes.

3

Mission Outcomes

Comprehensive outcome analysis per launch site provided insights into operational efficiency and success patterns.



Project Impact & Future Directions



Achieved Excellence

Successfully demonstrated comprehensive data science workflow from collection through predictive modelling with professional-grade visualisations.



Practical Applications

Models provide actionable insights for mission planning, risk assessment, and launch site optimization decisions.



Future Enhancement

Expanding datasets, refining algorithms, and incorporating additional mission parameters will further improve predictive accuracy.